FINAL CAPSTONE REPORT: SMART CITY TRAFFIC INTELLIGENCE ENGINE

Corridor Scope: Westbound I-94 Transit Corridor Analytics & AI Prediction

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Audience: Smart-City Mobility Management & Infrastructure Team

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1. Project Objective & Pipeline Methodology

This capstone project transforms a noisy, raw dataset containing 48,204 hours of traffic signals along westbound I-94 into an intelligent, enterprise-grade mobility management solution. The development lifecycle was executed across a reproducible multi-stage architecture:

A. Ingestion & Schema Enforcement (pipeline.py)

Raw fields were securely parsed using exception-trapped Pandas scripts. Features were verified against an immutable schema checklist, gracefully terminating executions upon structural layout mismatches.

B. Deduplication & Localized Cleaning

Purged duplicated rows and standardized categorical strings to eliminate text alignment variants caused by trailing whitespaces.

C. Context-Aware Month-by-Month Imputation

Hardware sensor dropouts logged at absolute zero (0 Kelvin / -273.15°C) and unphysical rainfall sensor breaks (9,831 mm/hr glitches) were repaired using localized monthly running medians. This preserved crucial seasonal weather dynamics rather than skewing the data with global dataset averages.

D. ML Feature Engineering (feature_engineering.py)

Generated binary weekend indicators and transformed clock hours into continuous circular wave coordinates using trigonometric sine and cosine waves. Continuous inputs were scaled via Z-score Standardization and Min-Max constraints, and weather categories were one-hot encoded.

2. Statistical Insights & Exploratory Findings

A. Bimodal Traffic Distribution

Descriptive statistics expose an overall traffic volume mean of 3,259.82 and a median of 3,380.0 vehicles/hour, with an exceptionally large Standard Deviation of 1,986.84 (61% of the mean). This quantifies a highly volatile bimodal daily profile: a high-density weekday commuter saturation mode peaking at 06:00–08:00 and 16:00–17:00 (frequently testing a maximum throughput ceiling of 7,280 vehicles/hour), and an off-peak night mode dropping below 1,000 vehicles/hour.

B. Weather Carrying Constraints

Joint probability and odds-ratio testing (Odds Ratio = 0.736) confirm that weather states and traffic volume are not independent. Overcast conditions (`Clouds`) sustain the highest mean volumes (3,618.45 vehicles/hour), while severe convective weather anomalies (`Squall`) trigger an immediate 43% collapse in lane carrying capacity (dropping down to 2,061.75 vehicles/hour) as commuters slow down or cancel non-essential trips.

C. Linear Independence of Temperature

Temperature displays a very weak positive association with traffic volume (r = 0.1323), proving day-to-day temperature movements are not a primary driver of volume. Peak traffic ceilings remain sticky across a massive spectrum from -15°C to +30°C.

3. Artificial Intelligence & Machine Learning Inference

A. Supervised Predictive Performance Matrix

To enable predictive risk analytics without an empirical accidents dataset, a data-driven binary classification target, high_risk, was synthesized. A specific timeline hour is flagged as high-risk (1) only when severe congestion (the top 25th percentile quartile) occurs simultaneously with hazardous precipitation or low-visibility atmospheric matrices.

Regression (Target: traffic_volume)

Baseline Linear Regression performed poorly (R^2 = 0.7257, MAE = 813.70 vehicles) due to its inability to map complex bimodal non-linear diurnal waves. The Random Forest Regressor achieved outstanding fit accuracy (R^2 = 0.9466, MAE = 266.92 vehicles), capturing complex temporal feature interactions cleanly.

Classification (Target: high_risk)

Baseline Logistic Regression yielded moderate stability (ROC AUC = 0.9920). The Random Forest Classifier achieved near-flawless execution boundaries (Accuracy = 98.25%, F1-Score = 0.9552, ROC AUC = 0.9974), precisely grouping safety risks.

B. Unsupervised & Deep Learning Sequential Architectures

K-Means Clustering

Grouped traffic conditions into 3 operational environments: Off-Peak Overnight Flow (Cluster 0: low volume, early morning), Daytime Core Transit Plateau (Cluster 1: normal volume, midday), and Peak Congestion & Weather Friction (Cluster 2: max volume, late afternoon storm states).

Apriori Association Rules

Mined patterns with high lift metrics. The rule matching Weekday Morning Peaks to Severe Congestion yielded a Lift of ~3.79, proving severe congestion is 3.79 times more likely to manifest under these combined conditions than random chance.

LSTM Deep Learning

A recurrent neural network utilizing a 24-hour lookback window processed sequential traffic behavior, achieving a strong continuous demand prediction matrix.

Model Explainability (XAI)

To overcome the complex hidden state layers of the LSTM, TreeSHAP was applied to a comparable surrogate tree ensemble. The SHAP summary plots visually proved that time-of-day coordinates ('Hour_Sin'/'Hour_Cos') and weekend distribution shifts hold the highest contribution weights over model output magnitude, while weather inputs remain minor background adjustments.

4. MLOps Simulation & Smart City Governance

The system transitions from local experimentation to an active smart-city architecture via three MLOps layers:

A. MLflow Tracking & Versioning (mlflow_model_registry.json)

Tracks model hyper-parameters and versions systematically, promoting `v2.0.0_Production_Champion` (Random Forest) as the live inference asset.

B. REST API Deployment (MLOps and Deployment Simulation.py)

Microservice endpoints build prediction lookups over live JSON data payloads ('Hour', 'Is_Weekend', 'Scaled_Temp'), capturing feature format anomalies gracefully.

C. Telemetry Drift Monitoring & Dashboard Alerting

Tracks streaming data real-time. If temperature features drift past 15K from the baseline or if prediction error spikes past a 1,000 MAE ceiling, the operations dashboard instantly switches from 'PASS / Normal' to a red 'ALERT / Requires Investigation' state, safely triggers a fallback system script, and alerts operators.

Conclusion: Sustainable AI Footprint Verdict

To support the city's environmental sustainability targets, the Random Forest Regressor was selected over the deep LSTM network for final production deployment. The massive computational GPU matrix overhead required to run a neural network 24/7 is not justified by the tiny accuracy gain over a tree-ensemble model that computes low-impact CPU decision splits instantly, matching municipal eco-design mandates perfectly.